

Municipal Infrastructure Damage Reporting App

ID	TYPE	DESCRIPTION	PRIORITY	ACCEPTENCE CRITERIA	RATIONALE	COMMENTS
FR-001	Functional	The system shall extract EXIF GPS metadata from an uploaded damage photo and use it to pinpoint the reported incident location on the municipal GIS map.	High	Pass: A photo containing valid GPS metadata is uploaded and the system displays the corresponding incident location on the municipal GIS map. Fail: The report is submitted without location data when GPS information is required.	Accurate location identification allows the municipality to determine where the infrastructure damage has occurred and route the report appropriately.	Clear and testable.
FR-002	Functional	The system shall allow the Citizen Reporter to submit a report for infrastructure damage such as a road pothole, broken streetlight, or water pipeline leak.	High	Pass: The citizen enters the required damage information and successfully submits the report. Fail: The system does not create a report when required information is missing.	The primary purpose of the application is to allow citizens to report municipal infrastructure damage.	Clear and testable.
FR-003	Functional	The system shall allow the Citizen Reporter to upload a photo showing the reported infrastructure damage.	High	Pass: A valid damage photo is uploaded and attached to the report. Fail: An invalid or unsupported photo cannot be attached to the report.	The problem statement requires citizens to report infrastructure damage using geotagged photos.	Clear and testable.

FR-004	Functional	The system shall automatically dispatch each submitted damage report to the relevant municipal engineering department.	High	Pass: A submitted pothole report is routed to the appropriate Road Infrastructure division or relevant municipal engineering department. Fail: A submitted report remains unassigned or is routed to an unrelated department.	Automated dispatch ensures that reported infrastructure damage reaches the appropriate municipal engineering department for action.	Clear and testable.
FR-005	Functional	The system shall allow the Municipal Engineer to review an assigned damage report and update its status.	Medium	Pass: The Municipal Engineer can view an assigned report and update its status successfully. Fail: The engineer cannot update the status of an assigned report.	Municipal engineers are responsible for handling the damage reports assigned to them.	Clear and testable.
NFR-001	Nonfunctional	The reporting portal shall support offline report drafting and automatically synchronize the drafted report when mobile network connectivity resumes.	High	Pass: A citizen can draft a damage report while offline, and the report is automatically synchronized when network connectivity is restored. Fail: The drafted report is lost or remains unsynchronized after connectivity is restored.	Ensures citizens can prepare damage reports even when mobile network connectivity is unavailable.	Clear and testable.
NFR-002	Nonfunctional	The system shall protect damage reports and uploaded photos from unauthorized access or modification.	High	Pass: Only authorized users can access or modify damage reports according to their roles. Fail: An unauthorized user can access or modify a damage report.	Protects citizen-submitted reports and photos from unauthorized access or modification.	Clear and testable.